Reg. No. | | | | | | | | | | |

[18MSCH3H02]

THIRD SEMESTER UNIVERSITY EXAMINATION, DECEMBER 2021

ORGANIC CHEMISTRY-III

Time : 03 Hours

Maximum Marks:70

Instructions:
1. Question 1 is compulsory, carries 10 marks.
2. Questions 2 to 7 carries 15 marks each, answer any FOUR.

1. Answer any FIVE of the following questions. 5 x 2 = 10

a. Explain catalytic hydrogenation.

b. What is functional group interconversion (FGI)? Give an example.

c. How Baker's yeast helps in ketone reduction?

d. Write a note on [1,3] dipolar cycloaddition reactions.

e. Write the structures of the following heterocycles;

i) isoquinoline; ii) indole; iii) benzimdiazole and iv) isoflavone.

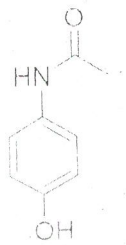
f. What is a hormone? Write the general structure of steroid.

2. Explain enantioselective and diastereoselective synthesis with appropriate example. (15)

3. a. Show the retrosynthetic analysis and synthetic method for saccharine.

b. Show the retrosynthesis of 3-nitrobenzaldehyde to get toluene as a starting material and discuss its synthesis.

c. Show the disconnection approach for the target molecule paracetamol (TM) to get phenol as one of the starting material and explain its synthesis.

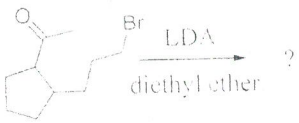


TM

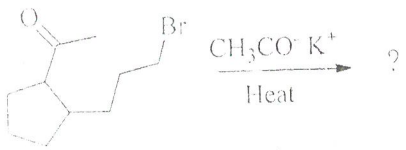
(5+5+5)

4. Predict the products for the following reactions and explain the mechanism;

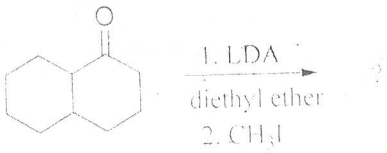
a)



b)



c)



(5+5+5)

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a. Geometry of ketenes allow a suprafacial [2+2] cycloaddition with alkenes in presence of photons, unlike in thermal conditions. Explain it by molecular orbital correlation.

b. What is Diels-Alder reaction? Give examples and describe the favorable conditions for dienes, and dienophile with regioselectivity. **(5+10)**

a. Write a note on synthesis of flavones using Baker-Venkataraman rearrangement.

b. Write the resonance structures of pyrimidine and explain its electrophilic substitution reactions.

c. Explain any two methods for the synthesis of benzothiazole. **(5+5+5)**

a. Give an account on the classification of carbohydrates with suitable examples.

b. Write a note on gestogens. **(10+5)**